

1.7 Equilibria

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.7 Equilibria
Difficulty	Medium

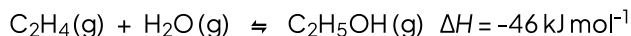
Time allowed: 80

Score: /60

Percentage: /100

Question 1a

Ethanol has a great number of uses. For industrial purposes, it can be manufactured via the following reversible reaction.



The optimum pressure for this reaction is between 60 and 70 atm.

State and explain the effect, if any, that increasing the overall pressure would have on the equilibrium yield of ethanol. Use Le Chatelier's principle to explain your answer.

[3 marks]

Question 1b

Although Le Chatelier's principle is used to suggest the best conditions for a reaction, often a compromise has to be made.

i)

Use Le Chatelier's principle to suggest whether a high or low temperature should be used to produce the maximum yield of ethanol in the reaction from part (a).

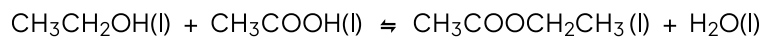
ii)

State a problem that may occur from using this temperature.

[4 marks]

Question 1c

Ethanol can be used as a reactant in another equilibrium reaction; the manufacture of ethyl ethanoate.



Give the expression for the equilibrium constant, K_c , for this equilibrium.

$$K_c =$$

[1 mark]

Question 1d

A student set up the esterification reaction seen in part (c), adding ethanol and ethanoic acid to a reaction vessel. They set the reaction up in a closed system, at a constant temperature and allowed equilibrium to be reached.

The reaction was done in a container with a volume of 250 cm^3 .

Table 1.1 below shows the amount of each substance present in the equilibrium mixture.

Table 1.1

Substance	Amount (mol)
$\text{CH}_3\text{CH}_2\text{OH}$	0.0375
CH_3COOH	0.0615
$\text{CH}_3\text{COOCH}_2\text{CH}_3$	0.0776
H_2O	0.0834

i)

Calculate K_c for this reaction to 2 decimal places.

$$K_c = \dots\dots\dots$$

[3]

ii)

Deduce the units for K_c .

$$\text{units} = \dots\dots\dots$$

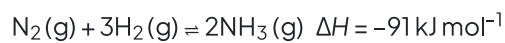
[1]

[4 marks]

Question 2a

Ammonia can be manufactured from nitrogen and hydrogen gases in the Haber process.

The equilibrium is shown below.



Write an expression for K_c for this equilibrium.

$K_c =$

[1 mark]

Question 2b

A chemist carries out a series of experiments to investigate the conversion of nitrogen and hydrogen into ammonia under different conditions.

The chemist mixes together 10.40 mol N_2 and 22.50 mol H_2 and pressurises the gases so that the total gas volume is 5.00 dm^3 .

The mixture is allowed to reach equilibrium at constant temperature and without changing the total gas volume. The equilibrium mixture contains 5.60 mol NH_3 .

i)

Calculate K_c at this temperature. Give your answer to three significant figures.

Show your working.

$K_c = \dots\dots\dots$

[5]

ii)

State the units for K_c .

units = $\dots\dots\dots$

[1]

[6 marks]

Question 2c

The chemist repeats the experiment several times. In each experiment, the chemist makes one change.

The chemist heats the mixture to a higher temperature at constant pressure.

Explain whether the value of K_c would be greater, smaller or the same.

[1 mark]

Question 2d

The chemist increases the pressure of the mixture at constant temperature.

d)

Explain whether the value of K_c would be greater, smaller or the same.

[1 mark]

Question 3a

Hydrogen chloride undergoes a reversible reaction with oxygen at 400 °C at in the presence of copper(II) chloride catalyst forming chlorine and water.

i)

Write the equation for this reaction. Include state symbols

[2]

ii)

The reaction exists in dynamic equilibrium. The reaction was repeated at the same temperature but at a lower pressure.

State and explain the effect on the composition of the equilibrium mixture of the change in pressure.

[2]

[4 marks]

Question 3b

When 2.00 mol of HCl are mixed in a sealed container with 0.600 mol of O_2 at 400 °C, 0.700 mol of Cl_2 and 0.700 mol of H_2O are formed.

The total pressure inside the container is 1.15×10^5 Pa.

i)

Calculate the amounts, in mol, of HCl and O_2 in the equilibrium mixture.

HCl = mol

O_2 = mol

[2]

ii)

Use your answer to part **(b)(i)** to calculate the mole fraction and hence partial pressure of each species in the equilibrium mixture.

p_{HCl} = Pa

p_{O_2} = Pa

p_{Cl_2} = Pa

p_{H_2O} = Pa

[3]

iii)

Write an expression for K_p for this reaction.

K_p =

[1]

iv)

Using your answer to part **(b)(ii)** and part **(b)(iii)** calculate a value for K_p and deduce the units.

K_p =

units =

[2]

[8 marks]

Question 3c

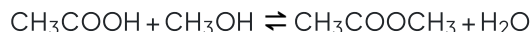
The reaction is repeated without a catalyst.

State the effect of this on K_p

[1 mark]

Question 4a

Ethanoic acid can be reacted with methanoic acid to form methyl ethanoate, and an equilibrium mixture is formed.



The reaction is usually carried out in the presence of an acid catalyst.

Write an expression for the equilibrium constant, K_c , for this reaction, clearly stating the units.

$K_c =$

units

[2 marks]

Question 4b

In an experiment to determine K_c a student placed together in a conical flask 0.20 mol of ethanoic acid, 0.20 mol of methanol, and 0.01 mol of hydrogen chloride catalyst.

The flask was sealed and kept at 25 °C for ten days. After this time, the student titrated all of the contents of the flask with 1.80 mol dm⁻³ KOH. At the end-point, 26.30 cm³ of KOH had been used.

i)
Write a balanced equation for the reaction between ethanoic acid and potassium hydroxide.

[1]

ii)
Suggest a suitable choice of indicator for the titration.

[1]

[2 marks]

Question 4c

The student carried out the following calculations.

i)

Calculate the amount, in moles, of KOH used in the titration.

[1]

ii)

What amount, in moles, of this KOH reacted with the hydrogen chloride?

[1]

iii)

Hence calculate the amount, in moles, of KOH that reacted with the ethanoic acid.

[1]

iv)

Use your results from parts **(i)** **(ii)** and **(iii)** to calculate the amount, in moles, of ethanoic acid present at equilibrium.

Hence complete the table below

[2]

	CH ₃ COOH	CH ₃ OH	CH ₃ COOCH ₃	H ₂ O
Initial amount / mol	0.20	0.20	0	0
Equilibrium amount / mol				

v)

Use your results to calculate a value for K_c for this reaction and state its units.

K_c =

units =

[2]

[7 marks]

Question 4d

The student performed another titration between the ethanoic acid used in the formation of the ester and the potassium hydroxide solution. The volume of potassium hydroxide added from the burette was recorded.

Sketch a pH titration curve for this reaction.

[4 marks]**Question 5a**

This question is about Brønsted-Lowry acids and bases.

i)
Give the meaning of the term Brønsted-Lowry base.

ii)
Explain the term weak acid.

[3 marks]

Question 5b

A student titrated 0.10 mol dm^{-3} acid into a conical flask containing 25.0 cm^3 of 0.1 mol dm^{-3} of a base, recording the pH with each addition of acid.

The student repeated the procedure using different combinations of acids and bases.

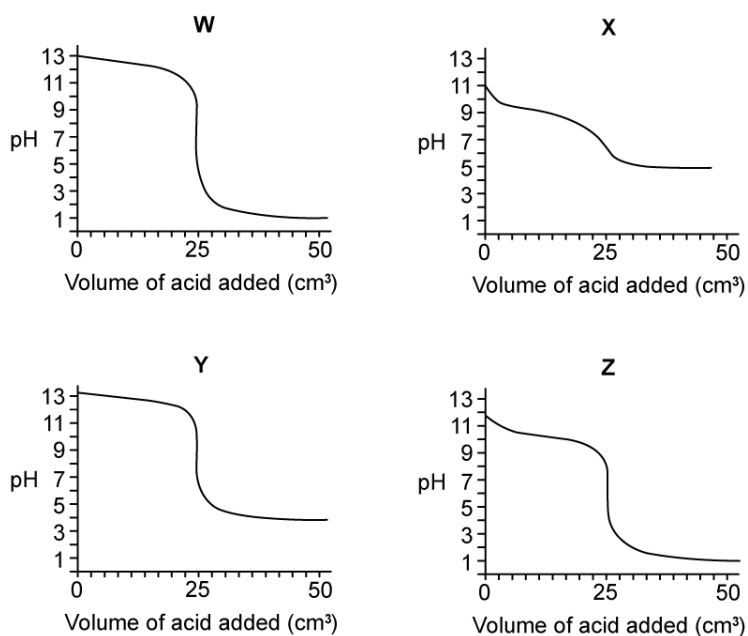


Fig. 5.1

Identify which curve shown above in Fig. 5.1, is the acid-base combination for the following:

- Ammonia and ethanoic acid.
- Ammonia and nitric acid.
- Sodium hydroxide and propanoic acid.

[3 marks]

Question 5c

Identify which indicator given in Table 5.2 would be most suitable for curve **Z**. Justify your answer.

Table 5.2

Indicator	pH range
Methyl orange	3.1 – 4.4
Phenolphthalein	8.3 – 10.0
Bromothymol blue	6.0 – 7.6

[3 marks]**Question 5d**

Water is amphoteric and will react with nitric acid and ammonia.

i)

Write an equation to show how water reacts with ammonia

[1]

ii)

Explain the term amphoteric.

[1]**[2 marks]**

Model Answers: Medium

1a

Increasing the overall pressure of this system would have the following effect on the equilibrium yield of ethanol:

- The yield of ethanol would increase; [1 mark]
- (According to Le Chatelier's principle) increasing the pressure shifts (the position of) equilibrium in the direction of / to the side with the fewest number of (gas) molecules

OR

There are fewer molecules (of gas) on the right hand side / fewer molecules of product;

OR

There are 2 molecules (of gas) of reactants and 1 molecule (of gas) of product; [1 mark]

- To oppose the change/increase in pressure

OR

To decrease the pressure; [1 mark]

[Total: 3 marks]

- Le Chatelier's principle states that if a system at equilibrium is disturbed, then the equilibrium moves in the direction that will reduce/counteract the disturbance
- **Watch out** - a change in pressure will only affect reactions involving gases
 - In some reactions, not all of the reactants and products in the equation will be gases
 - When deciding which direction the equilibrium position will shift in, **only count the molecules of gases**
- If you increase the pressure of a system, the position of equilibrium will move to counteract the change, and will move to 'the side' which has fewer molecules of gas.
- So, the position of equilibrium shifts to the right (fewer molecules of gas) which causes a greater yield of product (ethanol) to be produced
- If the pressure is decreased then the opposite happens - the position of equilibrium shifts to the side with the greatest number of molecules of gas

1b

b)

i) According to Le Chatelier's principle, the optimum temperature would be:

- Low (temperature); [1 mark]
- The forward reaction is exothermic / the forward reaction releases heat

OR

The enthalpy change for the (forward) reaction is $\Delta H = -46 \text{ kJ mol}^{-1}$

OR

The backward reaction is endothermic / the backward reaction absorbs heat; [1 mark]

- (At a lower temperature) the (position of) equilibrium would shift to the right hand side / towards the products / move to increase the temperature / release heat energy; [1 mark]

ii) A problem which may occur from using this temperature is:

- (A low temperature can result in a) slow rate of reaction

OR

Product would be produced too slowly; [1 mark]

[Total: 4 marks]

- Exothermic reactions **release heat energy** to the surroundings, whereas endothermic reactions **absorb heat energy**
 - The forward reaction in this question is exothermic, the backward reaction is endothermic
 - If the temperature of the system is **low/decreased**, the equilibrium will move in whichever direction will **oppose the change** and **'heat the reaction up'** – this is always the exothermic reaction direction, because it will release heat energy
 - If the temperature was higher, the opposite would happen and the position of equilibrium would shift to the left, meaning the yield of ethanol would decrease
- If something is underlined in a mark scheme then you **must** include it in your answer
 - There is a lot underlined in this mark scheme – this is because throughout this topic, you have to be specific and clear with your explanations
- Students lose marks on these questions because their answers do not clearly

explain what is happening to the position of equilibrium

1c

The K_c expression for this reaction is:

$$K_c = \frac{[\text{CH}_3\text{COOCH}_2\text{CH}_3][\text{H}_2\text{O}]}{[\text{CH}_3\text{CH}_2\text{OH}][\text{CH}_3\text{COOH}]}$$

- The expression should be drawn exactly as above - products on the top, reactants on the bottom, square brackets drawn throughout; [1 mark]

[Total: 1 mark]

- When writing the K_c expression, remember that:

$$K_c = \frac{[\text{Products}]}{[\text{Reactants}]}$$

- Square brackets represent “concentration of” - if you use circular brackets you will **not** be awarded the mark
- Unless specified, state symbols are not needed (they will be ignored if included)

1d

d)

i) To calculate K_c for the reaction:

- Convert all values in moles to concentrations; [1 mark]

$$\text{Concentration (mol dm}^{-3}\text{)} = \frac{\text{moles} \times 1000}{\text{volume (cm}^3\text{)}}$$

$$[\text{CH}_3\text{COOCH}_2\text{CH}_3] = \frac{0.0776 \times 1000}{250} = 0.3104$$

$$[\text{H}_2\text{O}] = \frac{0.0834 \times 1000}{250} = 0.3336$$

$$[\text{CH}_3\text{COOH}] = \frac{0.0615 \times 1000}{250} = 0.246$$

$$[\text{CH}_3\text{CH}_2\text{OH}] = \frac{0.0375 \times 1000}{250} = 0.15$$

OR

use concentration (mol dm^{-3}) = $\frac{\text{moles}}{\text{volume (dm}^3\text{)}}$ and divide moles by 0.250 for each;
[1 mark]

- Input values into K_c expression (allow if values from table inserted correctly); [1 mark]

$$K_c = \frac{(0.3104) \times (0.3336)}{(0.15) \times (0.246)}$$

- $K_c = 2.81$ (must be to 2 decimal places **AND** correctly rounded); [1 mark]

ii) The units for K_c are:

- There are **no units** for this K_c ; [1 mark]

[Total: 4 marks]

- The amount of each substance is given in moles – K_c is calculated from concentrations
- To convert a volume from $\text{cm}^3 \rightarrow \text{dm}^3$, divide by 1000
- Square brackets represent “concentration of” – if you use circular brackets you will **not** be awarded the mark
- For units, write out an expression replacing each concentration with mol dm^{-3} , and then cancel out the units where you can:

$$\text{Units} = \frac{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})} = \text{No units}$$

- You must write “no units” or you will not get the mark

2a

The K_c expression for this equilibrium would be:

- $K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$; [1 mark]

[Total: 1 mark]

2b

b)

i) K_c can be calculated as follows:

- moles of $\text{N}_2 = 10.4 - \left(\frac{5.6}{2}\right) = 7.6(0) \text{ mol}$; [1 mark]
- moles of $\text{H}_2 = 22.5 - (1.5 \times 5.6) = 14.1(0) \text{ mol}$; [1 mark]
- $[\text{N}_2] = \frac{7.60}{5.00} = 1.52 \text{ (mol dm}^{-3}\text{)}$

AND

$$[\text{H}_2] = \frac{14.1}{5.00} = 2.82 \text{ (mol dm}^{-3}\text{)}$$

AND

$$[\text{NH}_3] = \frac{5.60}{5.00} = 1.12 \text{ (mol dm}^{-3}\text{)}; [1 \text{ mark}]$$

- $K_c = \frac{(1.12)^2}{1.52 \times (2.82)^3}$; [1 mark]
- 0.0368; [1 mark]

ii) The units of K_c are:

- Units of $\text{dm}^6 \text{ mol}^{-2}$; [1 mark]

Answer must be correct to 3 decimal places

[Total: 6 marks]

- First you need to calculate the amounts of nitrogen and hydrogen present at equilibrium
 - For nitrogen you need to divide by 2 because the ratio $\text{N}_2 : \text{NH}_3$ from the balanced equation is 1:2 ($\frac{1}{2} = 0.5$, so multiply by 0.5 which is the same as dividing by 2)

- For hydrogen you need to multiply by 1.5 because the ratio of H_2 to NH_3 is 3:2 ($\frac{3}{2} = 1.5$)
- You then calculate the concentrations of each species at equilibrium (moles $\div$ 5)
- K_c is then calculated using the expression from the previous part of the question
 - Be careful you manipulate the units correctly as this is a common mistake and an easy mark to get
- For the final units, plug the unit of each species into the K_c expression:
 - $$\frac{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$$
- Two above the line cancel out with two below the line leaving:
 - $$\frac{1}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$$
- This then becomes
 - $$\frac{1}{\text{mol}^2 \text{ dm}^{-6}}$$

2c

ii) the units of K_c are.

- Units of $\text{dm}^6 \text{mol}^{-2}$; [1 mark]

Answer must be correct to 3 decimal places

[Total: 6 marks]

- First you need to calculate the amounts of nitrogen and hydrogen present at equilibrium
 - For nitrogen you need to divide by 2 because the ratio $\text{N}_2 : \text{NH}_3$ from the balanced equation is 1:2 ($\frac{1}{2} = 0.5$, so multiply by 0.5 which is the same as dividing by 2)
 - For hydrogen you need to multiply by 1.5 because the ratio of H_2 to NH_3 is 3:2 ($\frac{3}{2} = 1.5$)
- You then calculate the concentrations of each species at equilibrium (moles $\div$ 5)
- K_c is then calculated using the expression from the previous part of the question
 - Be careful you manipulate the units correctly as this is a common mistake and an easy mark to get
- For the final units, plug the unit of each species into the K_c expression:
 - $$\frac{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$$
- Two above the line cancel out with two below the line leaving:
 - $$\frac{1}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$$
- This then becomes
 - $$\frac{1}{\text{mol}^2 \text{dm}^{-6}}$$

2c

The value for K_c would be:

- Smaller

AND

Forward reaction is exothermic **OR** ΔH is negative; [1 mark]

[Total: 1 mark]

- For equilibrium constants
 - If the reaction is exothermic
 - An increase in temperature will cause the value of K_c to decrease

- If the reaction is endothermic
 - An decrease in temperature will cause the value of K_c to increase

2d

The value of K_c will be:

- The same

AND

K_c is temperature dependent/ only changed by temperature

OR

K_c is not changed by pressure; [1 mark]

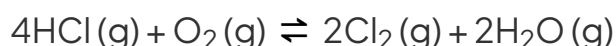
[Total: 1 mark]

- The equilibrium constant is unaffected by pressure, concentration or the presence of a catalyst
- It is only temperature dependent since the temperature influences the kinetic energy of the particles

3a

a)

i) The equation is:



- Correct balancing; [1 mark]
- Correct state symbols; [1 mark]

ii) The effect on the composition of the equilibrium mixture of the change in pressure is:

- Fewer moles of gas on RHS / 5 moles of gas on LHS and 4 moles of gas on RHS; [1 mark]
 - Equilibrium shifts to the LHS
- AND**
- Decreases yield of product; [1 mark]

[Total: 4 marks]

- You have been asked to add the state symbols for the reaction, at 400 °C the

reactants and products will all be gaseous

- To balance the equation
 - $\text{HCl(g)} + \text{O}_2\text{(g)} \rightleftharpoons \text{Cl}_2\text{(g)} + \text{H}_2\text{O(g)}$

Balance oxygen atoms

- $\text{HCl(g)} + \text{O}_2\text{(g)} \rightleftharpoons \text{Cl}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

Balance hydrogen atoms

- $4\text{HCl(g)} + \text{O}_2\text{(g)} \rightleftharpoons \text{Cl}_2\text{(g)} + 2\text{H}_2\text{O}$

Balance chlorine atoms

- $4\text{HCl(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{Cl}_2\text{(g)} + 2\text{H}_2\text{O}$

- Le Chatelier's principle states that if a system at equilibrium is disturbed, then the equilibrium moves in the direction that will reduce/counteract the disturbance
- If the pressure is decreased then the position of equilibrium shifts to the side with the greatest number of molecules of gas

3b

b)

i) The amounts, in mol, of each HCl and O₂ in the equilibrium mixture:

- Moles of HCl = 0.600 mol; [1 mark]
- Moles of O₂ = 0.250 mol; [1 mark]

ii) The partial pressure of each species in the equilibrium mixture:

- $p_{\text{HCl}} = 30\,705\text{ Pa}$; [1 mark]
- $p_{\text{O}_2} = 12\,765\text{ Pa}$; [1 mark]
- $p_{\text{Cl}_2} = 35\,765\text{ Pa}$

AND

$$p_{\text{H}_2\text{O}} = 35\,765\text{ Pa}; [1\text{ mark}]$$

iii) The K_p expression is:

$$K_p = \frac{(p_{\text{Cl}_2})^2 \times (p_{\text{H}_2\text{O}})^2}{(p_{\text{HCl}})^4 \times p_{\text{O}_2}}$$

OR

$$K_p = \frac{p_{\text{Cl}_2}^2 \times p_{\text{H}_2\text{O}}^2}{p_{\text{HCl}}^4 \times p_{\text{O}_2}}; [1 \text{ mark}]$$

iv) To calculate K_p and units:

- $K_p = \frac{(35\,765)^2 \times (35\,765)^2}{(30\,705)^4 \times 12\,765} = 1.44 \times 10^{-4}; [1 \text{ mark}]$
- Units = $\text{Pa}^{-1}; [1 \text{ mark}]$

[Total: 8 marks]

- In order to calculate the number of moles for HCl and Cl_2 at equilibrium as well as the partial pressure of each species you can use an ICE table
- Careful:** take note of the stoichiometry of the equation
 - To calculate the number of moles of HCl at equilibrium you need to use $2.00 - (2 \times)$ due to the 4:2 ratio
 - To calculate the number of moles of O_2 at equilibrium you need to $2.00 - (1/2 \times)$
- Mole fractions are calculated by dividing the moles of the individual species at equilibrium by the total number of moles = $0.600 + 0.250 + 0.700 + 0.700 = 2.250$ moles
- Partial pressures are calculated by multiplying the mole fraction by the total pressure

	HCl	O_2	Cl_2	H_2O
Initial number of moles	2.00	0.600	0	0
Eqm number of moles	$2.00 - (2 \times 0.700) = 0.600$	$0.600 - (1/2 \times 0.700) = 0.25$	0.700	0.700
Mole fraction	0.267	0.111	0.311	0.311
Partial pressure	$0.267 \times 1.15 \times 10^5$ = 30 705	$0.111 \times 1.15 \times 10^5$ = 12 765	$0.311 \times 1.15 \times 10^5$ = 35 765	$0.311 \times 1.15 \times 10^5$ = 35 765

- To determine the units write out the expression containing Pa
- Units = $\frac{\text{Pa}^2 \times \text{Pa}^2}{\text{Pa}^4 \times \text{Pa}} = \text{Pa}^{-1}$

3c

c) The effect of this on K_p is:

- K_p would not change; [1 mark]

[Total: 1 mark]

- Only temperature changes permanently affect the value of K_p

4a

a) The K_c expression and units are:

- $K_c = \frac{[\text{CH}_3\text{COOCH}_3][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{CH}_3\text{OH}]}$; [1 mark]
- No units; [1 mark]

[Total: 2 marks]

- **Remember:** For K_c expressions, the concentrations of the product go on the top, and the reactants on the bottom
- The unit for each substance will cancel each other out so there are no units for this K_c expression

4b

b)

i) The balanced equation for this reaction is:

- $\text{CH}_3\text{COOH} + \text{KOH} \rightarrow \text{CH}_3\text{CO}_2\text{K} + \text{H}_2\text{O}$; [1 mark]

ii) A suitable indicator is:

- Phenolphthalein; [1 mark]

[Total: 2 marks]

- The equation is a straightforward neutralisation reaction involving an acid and base which forms a salt and water
- A good indicator choice for a titration that involves a strong base is

phenolphthalein as it's pH range is above pH 7

4c

c)

i) The amount, in moles, of KOH used in the titration is:

- $1.80 \times 0.0263 = 0.04734$; [1 mark]

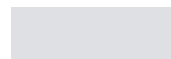
ii) The amount, in moles, of KOH that reacted with HCl is:

- moles KOH = moles HCl = 0.01 moles; [1 mark]

iii) The amount, in moles, of KOH that reacted with the ethanoic acid is:

- moles of KOH = $0.04734 - 0.01 = 0.03734$; [1 mark]

iv) The moles at equilibrium are:



- A good indicator choice for a titration that involves a strong base is phenolphthalein as its pH range is above pH 7

4c

c)

i) The amount, in moles, of KOH used in the titration is:

- $1.80 \times 0.0263 = 0.04734$; [1 mark]

ii) The amount, in moles, of KOH that reacted with HCl is:

- moles KOH = moles HCl = 0.01 moles; [1 mark]

iii) The amount, in moles, of KOH that reacted with the ethanoic acid is:

- moles of KOH = $0.04734 - 0.01 = 0.03734$; [1 mark]

iv) The moles at equilibrium are:

- Correct moles of CH_3COOH
AND
Correct moles of CH_3OH ; [1 mark]
- Correct moles of $\text{CH}_3\text{COOCH}_3$
AND
 H_2O ; [1 mark]

	CH_3COOH	CH_3OH	$\text{CH}_3\text{COOCH}_3$	H_2O
Initial amount / mol	0.20	0.20	0	0
Equilibrium amount / mol	0.16266	0.16266	0.03734	0.03734

v) A value for K_c and units are:

- $K_c = \frac{0.03734 \times 0.03734}{0.16266 \times 0.16266} = 0.0527$; [1 mark]
- No units; [1 mark]

[Total: 7 marks]

- Both neutralisation reactions involved have a 1:1 ratio

- $\text{KOH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOK} + \text{H}_2\text{O}$
- $\text{KOH} + \text{HCl} \rightarrow \text{KCl} + \text{H}_2\text{O}$
- Therefore the number of moles of potassium hydroxide will equal the number of moles of ethanoic acid and hydrogen chloride
- This question breaks down the steps for quite an in depth calculation
- To determine the moles in the equilibrium mixture you can use this table to help:

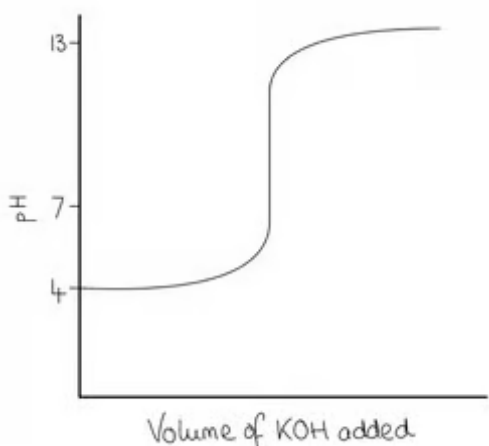
	CH_3COOH	CH_3OH	$\text{CH}_3\text{COOCH}_3$	H_2O
Initial amount / mol	0.20	0.20	0	0
Change	$0.20 - x$	$0.20 - x$	x	x
Equilibrium amount / mol	0.16266	0.16266	0.03734	0.03734

- $x = 0.03734$
- The units for K_c cancel out

4d

d) The titration curve should be:

- Alkali added; [1 mark]
- pH starts above pH 3; [1 mark]
- pH ends around pH 13; [1 mark]
- Equivalence point above pH 7; [1 mark]



[Total: 4 marks]

- **Careful:**

- A common mistake when drawing a pH curve is to add the wrong reactant, here the strong base is being added
- As it is a weak acid strong base titration the curve must start at the appropriate pH's
 - The curve must also start at a suitable pH for a weak acid
- The point of inflexion is where the line suddenly goes vertical
 - Halfway up this line is the equivalence point of the titration
 - For a weak acid – strong base titration this must be **above** pH 7

5a

a)

i) A Brønsted–Lowry base is:

- A proton acceptor; [1 mark]

ii) A weak acid is:

- A proton donor; [1 mark]
- Which only partially ionises / dissociates in water (to give H^+)

OR

Which does not fully ionise / dissociate in water (to give H^+); [1 mark]

[Total: 3 marks]

- These are very simple definitions, which get asked very often, so make sure you have learned them
- **Careful:** students often forget to give the second part of the definition for strong / weak **acid** – you need to remember to define the term acid as well

5b

b) The curves which represent each acid–base combination are:

i) Ammonia and ethanoic acid

- X; [1 mark]

ii) Ammonia and nitric acid

- Z; [1 mark]

iii) Sodium hydroxide and propanoic acid

- Y; [1 mark]

[Total: 3 marks]

- To help determine which curve represents which reaction it is useful to look at the pH value at the start and end of the reaction
 - This will give an indication if the acid/base is strong or weak.
 - In curve 'X' ammonia is a weak base and ethanoic acid is a weak acid and so the pH curve will start between pH 7–10 and finish between pH 4–7.
 - Nitric acid is a strong acid and so the curve will finish between pH 0–3 for curve 'Z'
 - As sodium hydroxide is a strong base, curve 'Y' will start with a high pH around pH 11–13 and as propanoic acid is a weak acid the curve will finish between pH 4–7

5c

c) The best indicator to use for curve **Z** and why is:

- Methyl orange; [1 mark]
- The curve shows strong acid – weak base; [1 mark]
- Equivalence point is below pH 7

OR

Methyl orange below pH 7; [1 mark]

[Total: 3 marks]

- The equivalence point is when the colour change occurs when the two solutions are mixed in equal proportions
- The pH range of bromothymol blue is too high for this titration

5d

d)

i) An equation to show how water reacts with ammonia is:



ii) Amphoteric means:

- Can behave as both an acid and base; [1 mark]

[Total: 2 marks]

- You should know what amphoteric means and be able to identify whether species are acting as a Brønsted-Lowry acid or base
- In this equation:
 - $\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{OH}^- + \text{NH}_4^+$ water is acting as a Brønsted-Lowry acid as it is accepting a proton

